

RESEARCH ARTICLE

Generalized graphical mixed models connect ecological theory with widely used statistical models

James T. Thorson 

Resource Ecology and Fisheries
Management Division, Alaska Fisheries
Science Center, National Marine Fisheries
Service, NOAA, Seattle, Washington, USA

Correspondence

James T. Thorson

Email: james.thorson@noaa.gov

Handling Editor: Nicolas Lecomte

Abstract

1. Ecological dynamics are often analysed across multiple sites, times, and variables. Ecologists typically represent interactions across space, time, and variables using generalized linear mixed models (GLMMs), generalized additive models (GAM), and structural equation models (SEM).
2. Here, I introduce the family of generalized graphical mixed models (GGMMs) and show that it extends GLMMs, GAMs, and SEMs. GGMMs represent ecological systems using a mathematical graph, where each analytic unit (node) has a direct effect on other units via specified linear interactions contained in a joint path matrix (edges). The GGMM first requires defining the dimensions and resolution of analytical units, and I illustrate concepts using combinations of sites, times, ages, and taxa. Next, the path matrix is constructed by combining elementary ecological relationships like ecological interactions, evolutionary trade-offs, time lags, and spatial diffusion. Finally, the GGMM is expressed as a simultaneous equation, efficiently estimated as a Gaussian Markov random field, and can be used for prediction, inference, and causal analysis.
3. I demonstrate GGMMs using three contrasting case studies and a simulation experiment, using a range of simple or complex software including an R-package (*tinyVAST*), base-R code for simulating dynamics, and bespoke estimation code using RTMB. Analysing age-structured population dynamics using *tinyVAST* shows that correlations by cohort can improve out-of-sample predictions. Simulating spatial dynamics in R shows that diffusive movement can be represented (or eliminated) in spatio-temporal models by including correlations across space, time, and a space–time interaction. Estimating separate Ornstein-Uhlenbeck parameters (OU) in a phylogenetic structural equation model using RTMB shows that geographic range has faster stabilizing selection than body size or specific metabolism across mammals. Finally, a simulation experiment confirms that the OU parameters can be better estimated while jointly interpolating missing data, which is not feasible using existing R-packages (*phylosem* or *phyloIm*).

This is an open access article under the terms of the [Creative Commons Attribution-NonCommercial-NoDerivs](https://creativecommons.org/licenses/by-nc-nd/4.0/) License, which permits use and distribution in any medium, provided the original work is properly cited, the use is non-commercial and no modifications or adaptations are made.

Published 2026. This article is a U.S. Government work and is in the public domain in the USA. *Methods in Ecology and Evolution* published by John Wiley & Sons Ltd on behalf of British Ecological Society.

4. I conclude that GGMMs connect ecological theory with statistical models that are applied for inference, prediction, and causal analysis throughout ecology. In particular, GGMMs allow analysts to sequentially add (and test) mechanisms during bespoke development of simulation or estimation models while also reverting to common statistical models when appropriate.

KEYWORDS

diffusion, generalized additive model, generalized linear mixed model, mathematical graph, phylogenetic path analysis, species interactions, structural equation model

1 | INTRODUCTION

Most ecological studies involve measuring, explaining, and predicting dynamics across multiple locations, times, and variables. For example, global conservation involves essential biodiversity variables (measurement of biomass across sites, times, and species), where statistical models are then used to fill in missing elements in this three-coordinate array (Jetz et al., 2019). Alternatively, macroevolution and paleoecology seek to identify how species traits change over time among species subject to extinction and speciation events (Hautmann, 2020). Ecological subfields generally differ in how they discretize space, time, and variables, but these ordinates remain ubiquitous across ecology.

Ecologists then use these measurements across space, time, and variables for three distinct tasks:

1. Inference, that is, measuring the functions and parameters that give rise to ecological dynamics in observational and experimental systems, so that these parameters can then be compared with ecological theory or other measured values across space and time.
2. Prediction, that is, estimating the value for a system variable where it was not specifically measured, for example, to allow biodiversity variables to be compared with target values.
3. Causal analysis, that is, estimating the value for a system variable under some hypothetical change, allows policy makers to compare the likely outcome of different potential management strategies and better understand mechanisms of the system.

These different tasks all involve some combination of experimental and observational studies, but an analysis suitable for one task may not be suitable for another (Levins, 1966). For example, a predictive model will often not be suitable for causal analysis (Arif & MacNeil, 2022), a model used to infer ecological parameters may explain a small proportion of predictive variance, and a model with good in-sample prediction may not provide any transferable inference about system parameters.

Given these contrasting goals, ecologists draw upon a large toolbox of mechanistic and statistical models. Here, I argue that most ecologists use some combination of generalized linear (mixed)

models (GLMM; Bolker et al., 2009), generalized additive models (GAM; Wood, 2017), and structural equation models (SEM; Grace, 2006) as their core toolbox for statistical analysis. For example, GAMs are widely used for estimating habitat utilization (Miller, 2025), GLMMs are used to control for pseudo-replication in the analysis of experimental designs (Bolker et al., 2009), and there is growing recognition that SEM (and variants like path analysis) have a distinct role for causal analysis (Grace, 2024). However, ecologists might ultimately seek to combine elements across these tools (e.g. incorporating random effects in causal analysis), and there is little conceptual guidance for how GLMM, GAM, and SEM models can be unified in a single theoretical framework.

I therefore introduce a family of generalized graphical mixed models (GGMMs) that connects ecological theory with these widely used statistical models. A GGMM starts by defining a mathematical 'graph', where any combination of variables, times, and sites are defined as nodes (visualized as boxes), and nodes are then linked using directed edges (visualized as arrows) that represent linear structural interactions. The edges are quantified using a sparse path matrix in a simultaneous equation, and developing a GGMM involves constructing this path matrix from elementary ecological processes including spatial diffusion, time lags, ecological interactions among species, or evolutionary trade-offs among traits. The resulting GGMM then includes GLMMs, GAMs, and SEM as nested submodels and is useful for inference, prediction, and causal analysis. I specifically emphasize two insights:

1. Graphs can be expressed using a sparse path matrix within a simultaneous equation and efficiently fitted as a Gaussian Markov random field;
2. Graphs can be assembled from elementary ecological relationships, such as time lags, spatial diffusion, evolutionary relatedness, and unstructured interactions. The joint path matrix is then computed by summing across the interaction (Kronecker product) of the path matrices resulting from these different elementary relationships.

I demonstrate these insights using three examples drawn from population dynamics, movement ecology, and macroevolution, and include a simulation experiment testing performance for the third example.

2 | MATERIALS AND METHODS

2.1 | Graphs, simultaneous equations, and random fields

In the following, I will construct a probabilistic graphical model by defining elementary graphs corresponding to basic ecological theories (each of which is represented by a path matrix), computing a joint path matrix by summing across the Kronecker product of these elementary path matrices, and then using that joint path matrix to define a GMRF for latent variables that is fitted as a generalized linear mixed model (GLMM). To explain the process in detail, I first introduce mathematical graphs including their construction, representation using simultaneous equations, and estimation using Gaussian Markov random fields (GMRFs), corresponding to Insight #1 from the introduction (see [Supporting Information 1](#) for more mathematical details).

As an illustrative example, I introduce the essential biogeographic variable $\mathbf{B}$ which contains biomass $b_{s,c,t}$ at each site $s \in \{1, 2, \dots, S\}$, species $c \in \{1, 2, \dots, C\}$, and time $t \in \{1, 2, \dots, T\}$. Given $S = 100$ sites, $C = 100$ species, and $T = 10$ times, we obtain an array $\mathbf{B}$ containing $STC = 100,000$ state-variables representing biomass for each site-species-time. Ecologists are then interested in interactions among species, which might include non-local properties (i.e. source-sink and predator avoidance behaviours) as well as both simultaneous and lagged effects (e.g. where predator c_1 in time t_1 consumes juvenile prey, and therefore has delayed impact on adult biomass for prey c_2 in time $t_2 = t_1 + \Delta_t$). In the limit, non-local and lagged interactions can result in approximately $(SCT)^2 / 2 = 5 \times 10^9$ interactions (excluding interactions backwards in time), so we obtain a staggering number of potential interactions even using this relatively coarse discretization of space, time, and taxonomy. I therefore seek some conceptually clear, computationally efficient, and statistically justified simplification for these non-local and lagged interactions.

A graphical model represents each element of state-variable $\mathbf{B}$ as a node (a.k.a. vertex, visualized as a box), and represents linear relationships as directed edges (visualized as arrows). This can then be represented as a simultaneous equation:

$$\begin{aligned} \text{vec}(\mathbf{B}) &= \mathbf{P}_{\text{joint}} \text{vec}(\mathbf{B}) + \text{vec}(\mathbf{E}) \\ \text{vec}(\mathbf{E}) &\sim \text{MVN}(\mathbf{0}, \mathbf{L}^T \mathbf{L}) \end{aligned} \quad (1)$$

where $\mathbf{P}_{\text{joint}}$ is the $K \times K$ 'joint' path matrix where K is the length of $\text{vec}(\mathbf{B})$, containing elements p_{k_2, k_1} that measure the effect of variable k_1 on k_2 (directed edges). Similarly, $\mathbf{E}$ is the array of exogenous variation representing processes that are not explicitly represented in the modelled interactions and $\mathbf{L}^T \mathbf{L}$ is the $K \times K$ covariance matrix for this unmodeled variation, represented using its square-root $\mathbf{L}$ containing λ_{k_2, k_1} (undirected edges). In its graphical representation, the joint path matrix $\mathbf{P}_{\text{joint}}$ can then be visualized as one-headed arrows where p_{k_2, k_1} points from k_1 to k_2 , and the exogenous variance $\mathbf{L}$ as two-headed arrows where λ_{k_2, k_1} connects

k_1 and k_2 . This 'box and arrow' visualization is widely used in path analysis (Wright, 1921), and I discuss the connections to causal modelling in a later section.

This simultaneous equation (and associated graphical representation) can be rearranged as a Gaussian Markov random field (GMRF; Rue & Held, 2005), where $\text{vec}(\mathbf{B})$ follows a multivariate normal distribution such that the inverse covariance ('precision') matrix can be constructed directly:

$$\begin{aligned} \text{vec}(\mathbf{B}) &\sim \text{MVN}(\mathbf{0}, \mathbf{Q}^{-1}) \\ \mathbf{Q} &= (\mathbf{I} - \mathbf{P}_{\text{joint}}^T) (\mathbf{L}^T \mathbf{L})^{-1} (\mathbf{I} - \mathbf{P}_{\text{joint}}) \end{aligned} \quad (2)$$

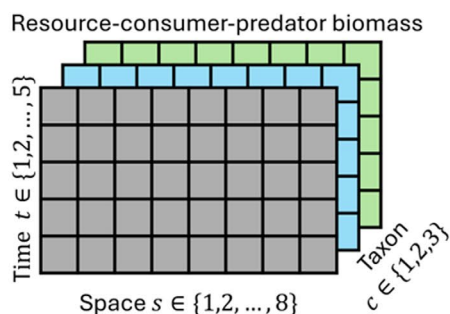
The probability density in the first line can be computed efficiently as long as joint path matrix $\mathbf{P}_{\text{joint}}$ and exogenous variation $\mathbf{L}$ are sparse, that is, species primarily interact with a constrained set of other species, within a localized neighbourhood, and where interactions occur simultaneously or over a reduced set of lags (see [Supporting Information 1](#) for more details). Although $\mathbf{Q}$ is often sparse, $\mathbf{Q}^{-1}$ will typically be dense, and therefore can represent covariance that propagates over large distances and/or time lags (e.g. first-order spatial diffusion results in an exponential correlation function across space, see Lindgren et al. (2011)). To identify parameters in $\mathbf{P}_{\text{joint}}$ and $\mathbf{L}$, we also require them to have constraints where for example, species interactions might be stationary across space and/or time. These constraints show up where element q_{k_2, k_1} of precision $\mathbf{Q}$ will be equal to q_{k_4, k_3} for other sites, times, or variables.

2.2 | Elementary graphs as building blocks

I next discuss how state-variables $\mathbf{B}$ can be defined, and the joint path matrix $\mathbf{P}_{\text{joint}}$ can then be constructed from elementary ecological relationships, such as time lags, spatial diffusion, evolutionary relatedness, and unstructured interactions (Insight #2 from the introduction; see [Figure 1](#)).

Constructing a graphical model first involves defining the state-variables and their resolution, which corresponds to the value recorded, dimensions, and length along each dimension of the state-variable $\mathbf{B}$. As discussed previously, $\mathbf{B}$ might be a three-dimensional array for community abundance (Space $\times$ Time $\times$ Taxa), where the analyst then decides how many species to include (or aggregate into functional groups) for Taxa, how many spatial jurisdictions or habitat patches to include (or how to discretize continuous spatial coordinates) for Space, and how to aggregate time intervals (e.g. quarters, years or decades) in Time. I demonstrate various two-dimensional problems in the following case studies, that is, the evolution of multiple traits (trait values across Taxa $\times$ Trait), diffusive animal movement (densities across Space $\times$ Time), and age-structured abundance (abundance across Age $\times$ Time), and use these varied examples to show how larger dimensions can be assembled from a small number of elementary processes.

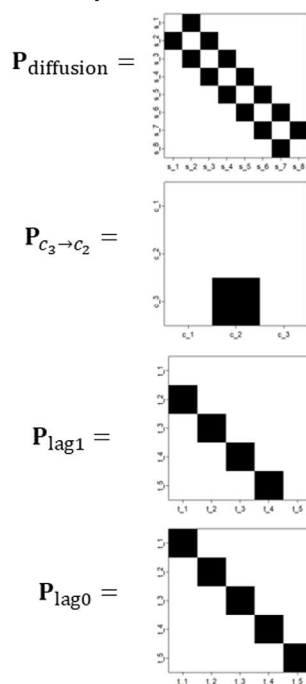
Step 1: Define dimensions for state-variable **B**



Step 2: Hypothesize mechanisms

Dynamics
 $\approx$
 Predator diffusion over time +
 Predator consumption of consumer +
 Predator density dependence +
 ...

Step 3: Construct elementary paths matrices



Step 4: Assemble joint path matrix

$$\mathbf{P}_{\text{joint}} = \begin{cases} \gamma_1 (\mathbf{P}_{\text{diffusion}} \otimes \mathbf{P}_{c_3 \rightarrow c_3} \otimes \mathbf{P}_{\text{lag1}}) + \\ \gamma_2 (\mathbf{I}_{\text{space}} \otimes \mathbf{P}_{c_3 \rightarrow c_2} \otimes \mathbf{P}_{\text{lag0}}) + \\ \gamma_3 (\mathbf{I}_{\text{space}} \otimes \mathbf{P}_{c_3 \rightarrow c_3} \otimes \mathbf{P}_{\text{lag1}}) + \\ \dots \end{cases}$$

Step 5: Evaluate probability of state-variable as Gaussian Markov random field

$$\text{vec}(\mathbf{B}) \sim \text{GMRF}(\mathbf{0}, \mathbf{Q})$$

$$\mathbf{Q} = (\mathbf{I} - \mathbf{P}_{\text{joint}}^T) \mathbf{V}^{-1} (\mathbf{I} - \mathbf{P}_{\text{joint}})$$

FIGURE 1 Schematic visualizing the workflow for specifying a generalized graphical mixed model for a hypothetical example involving three species (resource, consumer, and predator) over eight sites and five time intervals. This workflow involves: (1) defining the dimensions for the state-variable **B**; (2) hypothesizing mechanisms; (3) constructing elementary path matrices; (4) assembling the joint path matrix given hypotheses and their associated path matrices; and (5) evaluating the probability of the state-variable as a Gaussian Markov random field. See Figure 2 for more visualization regarding Step-3.

A graphical model can then be constructed by hypothesizing a set of elementary processes that specify a linear relationship among values (nodes) in response **B**. In the following, I focus on continuous response values (i.e. densities, continuous traits), and emphasize four elementary processes (see Figure 2), where each corresponds to a simultaneous equation and an associated path matrix **P** that is common in ecological analysis:

1. *Time-lagged dynamics*: Variable b_t can often be predicted from its first-order lag (b_{t-1}) and higher order lags (b_{t-2} , etc.) (Turchin, 2003). The first-order lag can be computed using lag-1 matrix $\mathbf{P}_{\text{lag1}}$, defined such that $\rho_{t_2, t_1} = 1$ if $t_2 = t_1 + 1$ and 0 otherwise (i.e. a sparse banded matrix). A lag-2 matrix is then calculated as $\mathbf{P}_{\text{lag2}} = \mathbf{P}_{\text{lag1}}^2$.
2. *Diffusive dynamics*: Ecological variables are often more similar when they are close together (Tobler, 1970), and this spatial autocorrelation can arise as animals or their physical habitat undergoes diffusion (Lindgren et al., 2011). In two dimensions and discretizing space into square grid cells, diffusion results in a location $\mathbf{s} = (x, y)$ affecting its four neighbours $\{(x, y + 1), (x + 1, y), (x, y - 1), (x - 1, y)\}$ such that each row of $\mathbf{P}_{\text{diffusion}}$ is non-zero for only four 'adjacent' elements, and $\mathbf{P}_{\text{diffusion}}$ then serves as the 'weight matrix' in a simultaneous autoregressive spatial model (Ver Hoef et al., 2018). Alternatively, metapopulation and metacommunity models often discretize

space into habitat patches (e.g. Hanski et al., 1994), and spatial adjacency and diffusion (and resulting spatial autocorrelation) can also be defined in this context;

3. *Evolutionary dynamics*: Ecologists often study evolutionary dynamics along a lineage (a phylogeny for species or a pedigree for individuals). A phylogeny is often represented as a tree, wherein a parent (individual or taxon) will split into two or more descendants and the evolutionary path matrix $\mathbf{P}_{\text{phylogeny}}$ is non-zero for each pair of descendant (row) and ancestor (column). Genetic drift occurring within a quadratic fitness landscape will result in a stabilizing selection towards the fitness peak (Lande, 1976);
4. *Interactive dynamics*: Finally, ecologists are often interested in structural linkages among variables. For example, a trophic cascade arises from two negative species interactions $\text{Predator} \rightarrow \text{Consumer}$ and $\text{Consumer} \rightarrow \text{Producer}$, where the product of these two negative direct effects results in a positive indirect effect from predators to producers. Interactions can then be used to construct the path matrix $\mathbf{P}_{\text{interaction}}$ with whatever pattern is hypothesized. Representing a species interaction as a linear effect (i.e. such that a change in predator density causes a $\rho_{\text{prey, predator}}$ impact on prey density) can be viewed as a first-order Taylor series approximation to a more general (e.g. Lotka-Volterra) function. Ecologists often represent interactions within smaller tightly-coupled 'species modules' (Holt, 1997), so representing species interactions will often result in $\mathbf{P}_{\text{interaction}}$ being sparse.

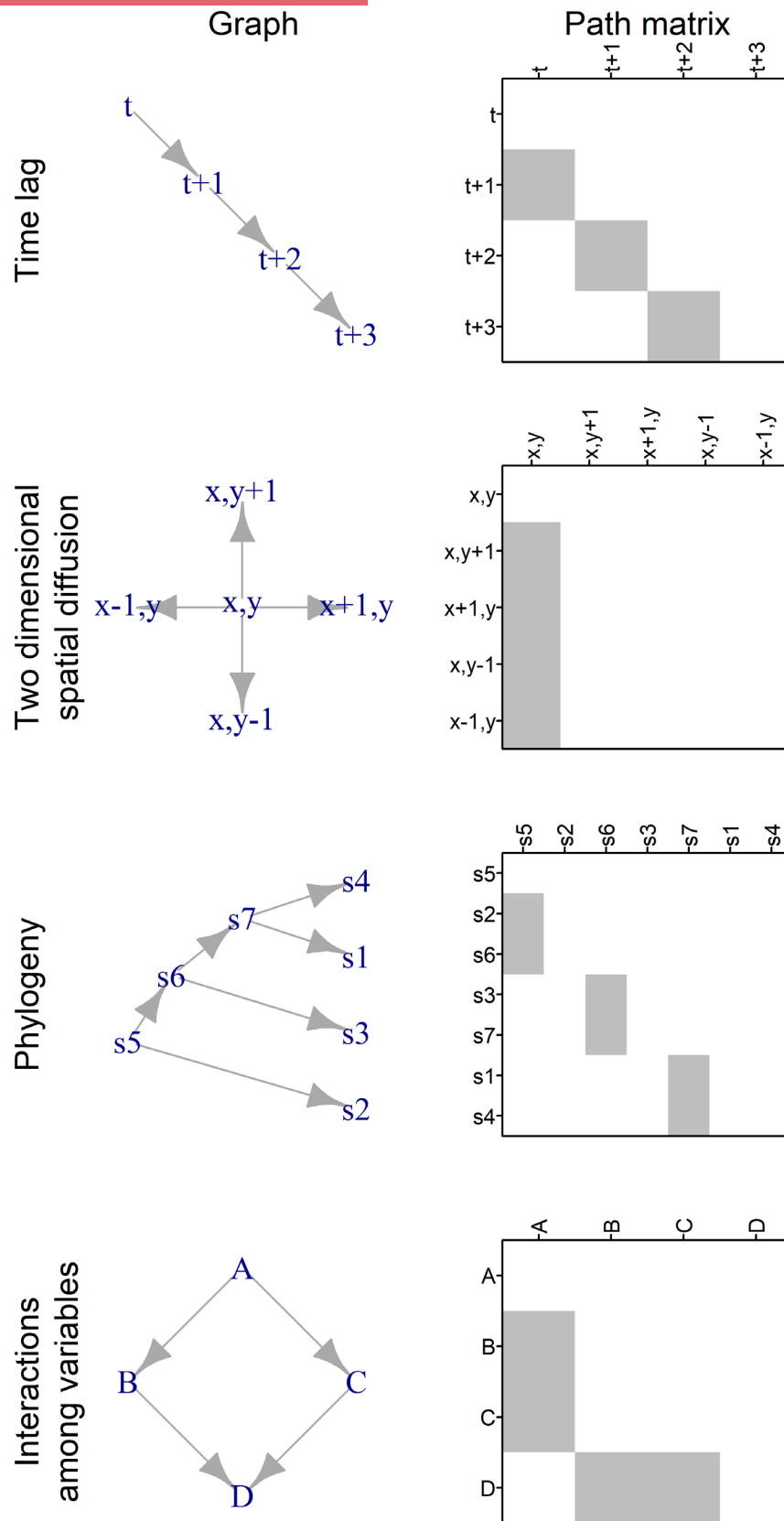


FIGURE 2 Graphs (left column) representing common simplifying assumptions for ecological dynamics, and the path matrix P (right column) in simultaneous equation $y = Py + \epsilon$ that results from each graph (with grey box when $p_{ij} \neq 0$ corresponding to graph arrows, and white boxes where $p_{ij} = 0$), showing first-order autoregressive dynamics (top row), spatial diffusion from a central location (x, y) when using square boxes to discretize a spatial domain in two dimensions with four adjacent grid cells (2nd row), a dated phylogeny (3rd row) showing ancestral nodes $\{s5, s6, s7\}$ and extant species $\{s1, s2, s3, s4\}$, and interactions among four variables $A \rightarrow B, A \rightarrow C, B \rightarrow D$, and $C \rightarrow D$ (4th row).

Other elementary graphs could also be developed for specific state-variables, e.g. a growth-transition matrix $\mathbf{P}_{\text{growth}}$ when state-variable $\mathbf{B}$ represents abundance by size and time.

These elementary relationships can then be combined to structure a larger multivariate model (see Figure 1). For example, an analyst might specify interactive dynamics where predator X affects prey Y and prey Y affects consumer Z (Graph-4 above) and where all taxa exhibit spatial diffusion (Graph-2). This involves two interactions $\rho_{X \rightarrow Y}$ and $\rho_{Y \rightarrow Z}$ in 3×3 matrices $\mathbf{P}_{XY}$ and $\mathbf{P}_{YZ}$ where $\mathbf{P}_{\text{interaction}} = \rho_{X \rightarrow Y} \mathbf{P}_{XY} + \rho_{Y \rightarrow Z} \mathbf{P}_{YZ}$, and an $S \times S$ matrix $\mathbf{P}_{\text{diffusion}}$ while estimating the strength of diffusion κ , where the joint path matrix is:

$$\mathbf{P}_{\text{joint}} = \underbrace{\rho_{XY} (\mathbf{P}_{XY} \otimes \kappa \mathbf{P}_{\text{diffusion}})}_{\text{diffusive effect of predator on consumer}} + \underbrace{\rho_{YZ} (\mathbf{P}_{YZ} \otimes \kappa \mathbf{P}_{\text{diffusion}})}_{\text{diffusive effect of consumer on producer}} \quad (3)$$

where $\mathbf{C} = \mathbf{A} \otimes \mathbf{B}$ is the Kronecker product of two matrices, such that resulting matrix $\mathbf{C}$ has dimensions $a_1 b_1 \times a_2 b_2$ when matrix $\mathbf{A}$ has dimension $a_1 \times a_2$ and matrix $\mathbf{B}$ has dimension $b_1 \times b_2$. Therefore, $\mathbf{P}_{\text{joint}}$ is the $3S \times 3S$ matrix arising from three parameters (two interactions and one diffusion rate), given that interaction matrix are 3×3 and $\mathbf{P}_{\text{diffusion}}$ has dimension $S \times S$. As shown in Equation (3), the joint path matrix is constructed via the sum of Kronecker products of the elementary path matrices (see general form in Supporting Information 1: Equation S1.14).

More generally, many ecological and evolutionary processes can be represented by combining time lags ($\mathbf{P}_{\text{lag1}}$), diffusive spatial dynamics ($\mathbf{P}_{\text{diffusion}}$), phylogenetic relationships ($\mathbf{P}_{\text{phylogeny}}$), and interactions ($\mathbf{P}_{\text{interaction}}$), beyond the illustrative examples we have discussed so far. For example, combining species interactions with species-specific diffusion rates (e.g. extending Equation 3) results in emergent spatial patterns wherein species are spatially clustered even in a homogenous environment (Levin, 1974; Turing, 1952), and can therefore represent a 'neutral model' for spatial patterns in meta-community theory (Leibold et al., 2004). Similarly, vector autoregressive (VAR) models are constructed by combining time lags and species interactions (e.g. Thorson et al., 2024), and have been widely used to predict how species interactions affect community stability (Ives et al., 2003; Wootton & Emmerson, 2005). Past studies have also extended VAR models to represent spatial autocorrelation (i.e. combining time lags, interactions, and diffusion), and these spatial VARs have been used to study the community impact of invasive insects or recovering predators (Schliep et al., 2018; Thorson et al., 2017). Representing community dynamics as a network across space and species using phylogenetic and trait similarity remains an active area of research (García-Callejas et al., 2025).

To further illustrate, I next introduce how these simultaneous equations (and associated graphs) arise in three varied ecological examples (Thorson, 2026). These examples represent different sub-fields (population dynamics, spatial ecology, and macroevolution), but are also organized to proceed from relatively simple code (an R-package *tinyVAST* for estimation), to intermediate (bespoke simulation code in base-R), and then to full complexity (bespoke estimation code using RTMB). Ultimately, I believe that ecologists are

increasingly familiar with developing bespoke code using high-level code such as JAGS, STAN, or NIMBLE (Kéry & Kellner, 2024), so I recommend greater emphasis on developing bespoke GGMMs via a progression similar to that presented here.

2.3 | Case study 1: Tracking cohorts in age-structured demographics

As a first example, ecologists are often interested in predicting abundance at age $n_{a,t}$ for A ages and T years, which arises via survival from the preceding age and year $n_{a-1,t-1}$. However, $n_{a,t}$ might vary for all ages in a single year and therefore be predicted from $n_{a-1,t}$, or it might be affected by changes in survey availability for an age that is consistent across years (i.e. predictable from $n_{a,t-1}$). I therefore explore a model with three interactions, arising from a lag-1 process among years and a separate lag-1 process among ages which I call $\mathbf{G}_{\text{Year}}$ and $\mathbf{G}_{\text{Age}}$, respectively, to distinguish the two versions of the lag-1 matrix $\mathbf{P}_{\text{lag1}}$:

$$\mathbf{P}_{\text{joint}} = \underbrace{\rho_1 (\mathbf{G}_{\text{Age}} \otimes \mathbf{I}_{\text{Year}})}_{n_{a-1,t} \rightarrow n_{a,t}} + \underbrace{\rho_2 (\mathbf{I}_{\text{Age}} \otimes \mathbf{G}_{\text{Year}})}_{n_{a,t-1} \rightarrow n_{a,t}} + \underbrace{\rho_3 (\mathbf{G}_{\text{Age}} \otimes \mathbf{G}_{\text{Year}})}_{n_{a-1,t-1} \rightarrow n_{a,t}} \quad (4)$$

where $\mathbf{G}_{\text{Age}}$ is the $A \times A$ lag-1 matrix $\mathbf{P}_{\text{lag1}}$ among A ages, $\mathbf{I}_{\text{Age}}$ is the $A \times A$ identity matrix, and $\mathbf{G}_{\text{Year}}$ and $\mathbf{I}_{\text{Year}}$ are the corresponding $T \times T$ lag-1 and identity matrices across years (see Supporting Information 2 for more details). I fit this model to proportional abundance-at-age for rex sole in the Gulf of Alaska, which was sampled intermittently from 1992 to 2022 (McGilliard, 2024). I specify a log-linked Tweedie distribution (Foster & Bravington, 2013) for measurement errors, which provides a convenient distribution for compositional data in bottom trawl surveys (Thorson, Monnahan, & Hulson, 2023), and fit the model using package *tinyVAST* (Thorson et al., 2025) release 1.4.0 in the R statistical environment, which uses *TMB* and the Laplace approximation (Kristensen et al., 2016; Skaug & Fournier, 2006) to marginalize across unmeasured state-variables. I then use 10-fold cross-validation to compare parsimony among the eight models arising from estimating or fixing at zero the three parameters $\{\rho_1, \rho_2, \rho_3\}$ from Equation 4, which represent the relative importance of within-cohort, within-year, and within-age drivers for observed abundance-at-age (see Data accessibility statement to access the R-script).

2.4 | Example 2: Diffusion-enhanced spatio-temporal models

Ecologists have a long-running interest in spatial ecology including the trajectory and rate of expanding range edges for invasive species. Spatial statisticians have developed non-separable spatio-temporal models that incorporate diffusive dynamics (Lindgren et al., 2023), but these see little use in ecology to date. Here, I present a novel demonstration that graphical models can represent both separable and diffusion-enhanced spatio-temporal dynamics using an additive

path matrix, resulting from a lag-1 matrix $\mathbf{P}_{\text{lag1}}$ in time and a spatial diffusion matrix $\mathbf{P}_{\text{diffusion}}$:

$$\mathbf{P}_{\text{joint}} = \underbrace{\rho_1 (\mathbf{I}_{\text{time}} \otimes \mathbf{P}_{\text{diffusion}})}_{d_{s,t} \rightarrow d_{s,t+1}} + \underbrace{\rho_2 (\mathbf{P}_{\text{lag1}} \otimes \mathbf{I}_{\text{space}})}_{d_{s,t} \rightarrow d_{s+1,t}} + \underbrace{\rho_3 (\mathbf{P}_{\text{lag1}} \otimes \mathbf{P}_{\text{diffusion}})}_{d_{s,t} \rightarrow d_{s+1,t+1}} \quad (5)$$

I demonstrate this using a deterministic simulation by visualizing the density matrix $\mathbf{D}$ resulting from diffusive dynamics $\text{vec}(\mathbf{D}) = (\mathbf{I} - \mathbf{P}_{\text{joint}})^{-1} \text{vec}(\mathbf{E})$, where $\text{vec}(\mathbf{E})$ is an indicator vector such that $\mathbf{E}_{s_{\text{center}},1} = 1$ for location s_{center} at the centre of the spatial domain in time $t = 1$ and $\mathbf{E}_{s,t} = 0$ elsewhere. I visualize this diffusive process over a square spatial domain discretized into 21 rows and 21 columns ($S = 441$ square grid cells) and $T = 3$ times, while fixing $\rho_1 = 0.8$ and $\rho_2 = 0.1$, and varying the value of ρ_3 . Diffusive dynamics are expected to result in a linear increase in the mean-square displacement for the utilization distribution over time (see [Supporting Information 3](#) for more details). I demonstrate this as a simulation experiment (not conditioned upon fitting data) using R code, including the *Matrix* package (Bates et al., 2023) to interface with Eigen (Guennebaud & Jacob, 2010) for efficient computation involving sparse matrices (see Data accessibility statement).

2.5 | Case study 3: Phylogenetic trait imputation with varying stabilization rates

Ecologists also study how traits covary among natural populations, seeking to identify evolutionary trade-offs that arise from adaptation to shared fitness constraints. Recent research has developed phylogenetic SEM from the Kronecker product of trait-interactions and a single evolutionary matrix that is shared across traits (Thorson, Maureaud, et al., 2023; Thorson & van der Bijl, 2023), but this does not allow different traits to have different rates of stabilizing selection. I therefore present a novel extension, where I calculate the joint precision from a simultaneous equation that includes phylogenetic path matrix $\mathbf{P}_{\text{phylogeny}}$ and an interaction matrix $\mathbf{P}_{\text{interaction}}$. This results in joint precision:

$$\mathbf{Q}_{\text{joint}} = (\mathbf{I} - \mathbf{P}_{\text{joint}}^T) \mathbf{Q}_{\text{phylogeny}} (\mathbf{I} - \mathbf{P}_{\text{joint}}) \quad (6)$$

where $\mathbf{P}_{\text{joint}} = \mathbf{I} \otimes \mathbf{P}_{\text{interaction}}$, $\mathbf{P}_{\text{interaction}}$ is the $C \times C$ matrix of interactions among traits, and $\mathbf{I}$ is the $S \times S$ identity matrix, and $\mathbf{Q}_{\text{phylogeny}}$ is the block-diagonal matrix of evolutionary precisions for each trait (see [Supporting Information 4](#) for more details).

To illustrate, I download three traits from PanTHERIA (Jones et al., 2009), representing specific metabolic rate ($\text{mL O}_2/\text{g}$), adult body mass (g), and home range size (km^2). Body size has the highest proportion of data (3340 measurements), while other traits have fewer measurements ([Supporting Information 5, Table S1](#)). I specify a phylogenetic SEM with two interactions $\log(\text{size}) \rightarrow \log(\text{metabolism})$ and $\log(\text{size}) \rightarrow \log(\text{range})$. I also estimate the Ornstein-Uhlenbeck (OU) parameter θ_c for each trait, used to calculate:

$$\mathbf{Q}_{\text{phylogeny}} = \begin{bmatrix} \mathbf{Q}_1 & 0 & 0 \\ 0 & \mathbf{Q}_2 & 0 \\ 0 & 0 & \mathbf{Q}_3 \end{bmatrix} \quad (7)$$

where $\mathbf{Q}_1$, $\mathbf{Q}_2$, and $\mathbf{Q}_3$ are the evolutionary precisions matrices given OU parameters θ_1 , θ_2 , and θ_3 for log-size, log-metabolism, and log-range respectively. I fit this model using a dated phylogeny across 5911 mammal species and 185 million years of evolutionary history (Upham et al., 2019), where $\mathbf{P}_{\text{phylogeny},j}$ is the 11821×11821 matrix across all tips and ancestral nodes. I specifically fit the model using bespoke code in RTMB (Kristensen, 2024), which allows analysts to write bespoke GGMMs by defining elementary graphs in R for efficient computation involving sparse matrices (see Data accessibility statement for more details to access code). I then record the estimated OU parameters (θ_1 , θ_2 , and θ_3 and associated standard errors), which represent the strength of stabilizing selection for each trait (Lande, 1976).

2.6 | Simulation experiment

Finally, I provide a simulation experiment to demonstrate how GGMMs compare with existing ecological software, specifically emphasizing how they can (1) represent ecological mechanisms flexibility while (2) usefully propagating uncertainty. To do so, I use the fitted model from Case Study #3 and simulate 200 replicated data sets from each of three scenarios, representing high, medium, or low data availability (i.e. where each trait is independently missing measurements for 50%, 70%, or 90% of species). For each of 600 simulated data sets, I then fit the GGMM, and compare it with either:

1. *Phylogenetic linear models*: fitting three PLM using R-package *phylolm* (Tung Ho & Ané, 2014), either treating log-body size as response with no predictor ($\log(\text{size}) \sim 1$), log-metabolism as response with log-size as predictor ($\log(\text{metabolism}) \sim \log(\text{size})$), or log-area as response with log-size as predictor ($\log(\text{area}) \sim \log(\text{size})$). However, *phylolm* cannot jointly conduct inference (estimate parameters) and prediction (impute missing values), so each model is fitted only to those species that have all relevant traits measured (a 'pairwise complete data'). For each model, I then record the estimated OU parameter.
2. *Phylogenetic structural equation models*: fitting a single phylogenetic SEM using R-package *phylosem* (Thorson & van der Bijl, 2023), and using the same path diagram as the GGMM. However, *phylosem* is restricted to assuming a single OU parameter across all three traits (a 'separable' precision matrix). I therefore record the imputed value for traits.

I then compare the error in estimated OU parameters between the GGMM and *phylolm*, and the imputed traits between the GGMM and *phylosem*. Finally, I also calculate the z-scored error from GGMM

(i.e. $z = \frac{\hat{\theta} - \theta}{\text{SE}(\hat{\theta})}$ where $\hat{\theta}$ is the estimated value, θ the simulated (true) value, and $\text{SE}(\hat{\theta})$ the estimated standard error), where a well-performing estimator will have z-scored errors that (approximately) follow a normal distribution.

3 | RESULTS

In the GGMM fitting to proportional abundance-at-age for rex sole in the Gulf of Alaska using an R-package (*tinyVAST*), 10-fold cross-validation indicates that the model with interactions along cohorts ($\rho_1 = n_{a,t} \rightarrow n_{a+1,t+1}$) and along years ($\rho_2 = n_{a,t} \rightarrow n_{a,t+1}$) has lowest predictive error (Supporting Information 5, Table S2). The fitted model (Figure 3) estimates that cohort effects ($\rho_1 = 0.73$) are substantially

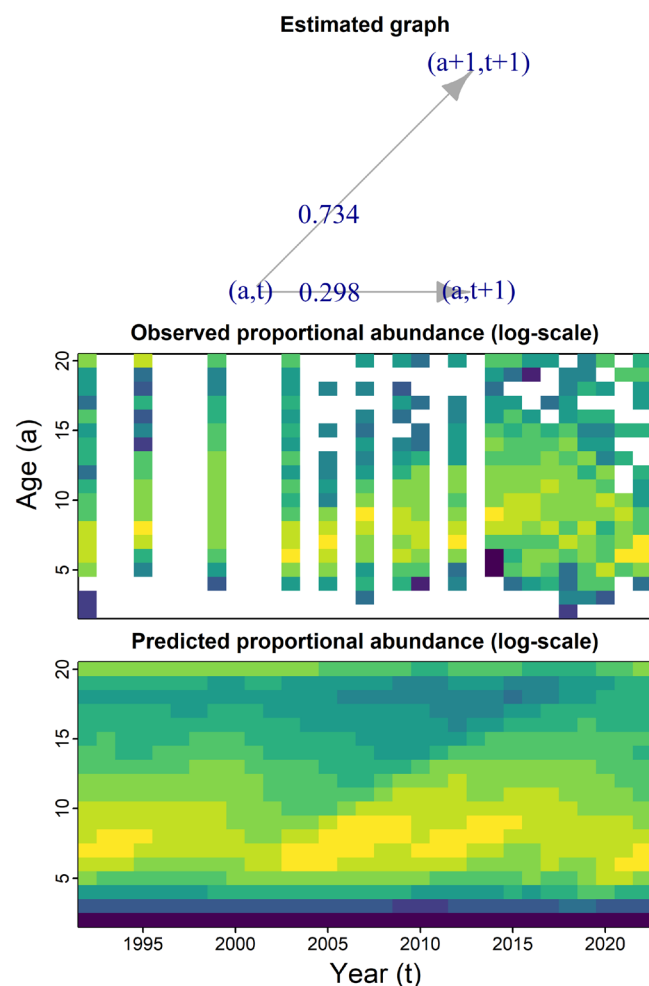


FIGURE 3 The estimated interactions (top panel) when predicting proportional abundance at age $n_{a,t}$ for rex sole in the Gulf of Alaska showing the estimated effect of survival along a cohort ρ_3 and effects along a year ρ_2 (see Equation 4), as well as the observed $n_{a,t}$ (middle panel) for each year (x-axis) and age (y-axis) showing low (purple) to high (yellow) values, and the estimated $n_{a,t}$ (bottom panel) including the estimated value for years with no direct sampling (white spaces in the middle panel).

stronger than year effects ($\rho_2 = 0.3$), and strong cohorts are also visually apparent starting at age-7 around 2005 and again in 2011, and progress visually through subsequent ages and years. Leave-year-out cross-validation (Supporting Information 5, Figure S1) confirms that including these interactions can result in skillful predictions for years without direct measurements.

The GGMM simulating diffusion-enhanced spatio-temporal dynamics in R (Figure 4) shows that diffusion arises by combining interactions across time ($n_{s,t} \rightarrow n_{s,t+1}$ using ρ_{time}) and interactions across spatial neighbours ($n_{s,t} \rightarrow n_{s+1,t}$ using ρ_{space}) (Figure 4 top row). In this scenario, mean-squared displacement (MSD) shows a close-to-linear increases over time with rate ρ_{space} , as expected given diffusive dynamics (where the departure from a linear increase in MSD arises from spatial boundary effects). The model then reverts to separable spatio-temporal dynamics when adding a parameter lagged spatial effect ($n_{s,t} \rightarrow n_{s+1,t+1}$ using $\rho_{\text{spacetime}}$) and fixing $\rho_{\text{spacetime}} = -\rho_{\text{space}}\rho_{\text{time}}$ (Figure 4 bottom row). Finally, intermediate dynamics arise when $-\rho_{\text{space}}\rho_{\text{time}} < \rho_{\text{spacetime}} < 0$. For example, when $\rho_{\text{spacetime}} = -0.5\rho_{\text{space}}\rho_{\text{time}}$ the MSD starts at 0.1 but then increases 0.05 per time interval (Figure 4 middle row). Importantly, $\rho_{\text{spacetime}}$ allows for a continuous bridge between two ecological hypotheses, where a hotspot remains stationary or diffuses outwards over time.

The GGMM using RTMB to estimate interactions among adult body size [$\ln(g)$], specific metabolic rate [$\ln(\text{mL } \frac{\text{O}_2}{g})$], and range size [$\ln(\text{km}^2)$] for 5911 mammal species over 185 million years (Figure 5) estimates a isometric ($\rho = 1.00$) scaling of range size with adult body size, and an allometric ($\rho = 0.69$) scaling of metabolic rate with body size. Additionally, it estimates weakest stabilizing selection for body size, with a nearly 20% correlation between two species separated by 150 million years of divergent evolution. By contrast, range size has strongest stabilizing selection (conditional upon body size), where a 20% correlation arises at approximately 25 million years of divergence. Finally, specific metabolism has an intermediate strength for stabilizing selection (20% correlation at 60 million years), conditional upon body size.

Finally, the simulation experiment shows that the GGMM and PLM both provide (approximately) unbiased estimates of the log-OU parameters (representing the strength of stabilizing selection) for all three traits (Figure 6 top row, where error distributions are centred on zero). However, the PLM cannot jointly estimate parameters and impute missing traits, and therefore is restricted to pairwise complete data. As a result, it has much larger errors (i.e. wider intervals) for the strength of stabilizing selection for specific metabolism and range size when 70%–90% of taxa are missing trait measurements. Alternatively, the GGMM and phylogenetic structural equation model (PSEM) have very similar levels of error when imputing missing traits, suggesting that pooling the OU parameter across traits has little impact on imputed trait estimates. Finally, the z-scored errors in estimated log-OU parameters (Figure S2) confirms that the standard errors provided by GGMM are generally reasonable (i.e. the simulation distribution matches the desired normal distribution). However, there appears to be some positive skewness for the simulation distribution (e.g.

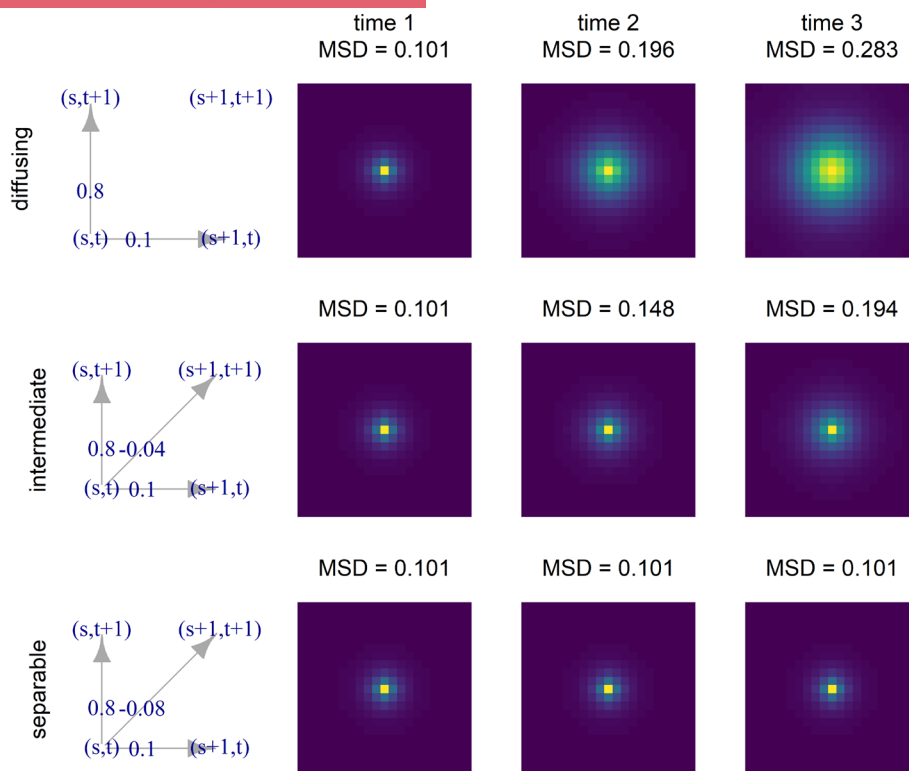


FIGURE 4 Visualizing diffusion-enhanced spatio-temporal dynamics (top row), intermediate dynamics (middle row), and non-diffusive (separable) spatio-temporal dynamics (bottom row), including the graph (1st column) linking a focal cell (s, t) and adjacent cells in a given time $(s + 1, t)$, the same cell in the next time $(s, t + 1)$, or adjacent cells in the next time $(s + 1, t + 1)$. Separable dynamics arise when $\rho_{s+1,t+1} = -\rho_{s,t+1}\rho_{s+1,t}$ (bottom row). I also visualize resulting dynamics from a concentrated density (purple is 0 density, yellow is high density) in time $t = 1$ (2nd column), and how this density evolves in times 2 (3rd column) and 3 (4th column). For each panel, I also calculate the mean-squared displacement (MSD) as the variance of the density function. Diffusive dynamics results in MSD increasing linearly over time, although the diffusive MSD in time-3 is slightly lower due to boundary effects.

bottom-right panel of Figure S2), suggesting that a likelihood profile might be more appropriate than assuming that errors are normally distributed.

4 | DISCUSSION

I introduced the family of GGMMs, which represent variables as nodes and interactions as directed edges and are efficiently fitted as a GMRF. In particular, ecological variables are often indexed by space, time, and category (e.g. species or age). Ecological interactions in GGMMs can be specified by combining several elementary graphical structures representing time lags, phylogenetic relatedness, spatial diffusion, ecological interactions among species, or evolutionary trade-offs among species traits. I then construct the joint path matrix for the GMRF from the sum of Kronecker products of the path matrices for elementary graphs. Using three varied case studies, I specifically showed that interactions among elementary graphical structures can represent population dynamics (age structure), movement dynamics (diffusion-enhanced spatio-temporal variation), and evolutionary dynamics (stabilizing selection among traits). This then yielded novel insights, for example, that stabilizing selection is stronger for adult home range than body

size within the mammal lineage. Finally, a simulation experiment confirmed that the GGMM can improve upon existing software (e.g. phylogenetic linear and SEM) by jointly imputing missing traits and estimating separate stabilizing selection for each trait, and that it can usefully propagate estimation uncertainty.

As shown in our second case study (diffusion-enhanced spatio-temporal models), GGMMs can be used as a spatial smoother similar to GAMs. In particular, the diffusion-enhanced spatio-temporal model constructs a joint precision matrix from structural parameters representing spatial diffusion, temporal autocorrelation, and the space-time interaction (see Figure 4). This precision matrix then serves a similar function to the penalty matrix that is used to minimize 'wiggleness' in a GAM (Miller, 2025). However, the GGMM has two added benefits relative to a GAM: (1) it derives the precision matrix from ecological interpretable parameters where for example, the space-time interaction parameter ρ_3 determines whether spatial hotspots propagate outwards in space over time; and (2) instead of using the path matrix to construct a precision (penalty) matrix, the path matrix can be repurposed to represent how animal diffusion affects, for example, the association between animal densities and habitat (Lindmark et al., 2026). Deriving the joint path matrix from ecological mechanisms and using this to define a penalty matrix therefore reveals a rich class of new dynamics for ecological

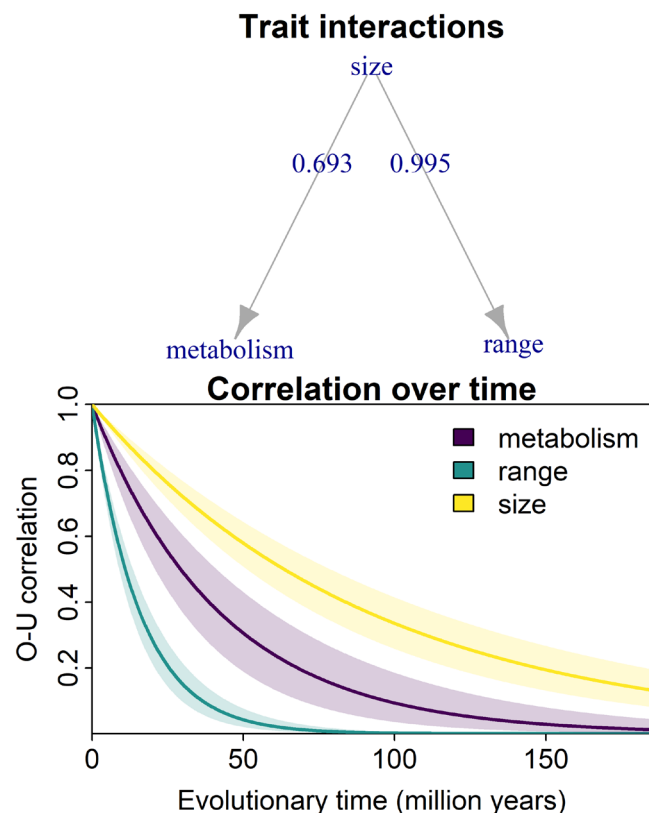


FIGURE 5 The estimated interaction (top panel) among three log-transformed traits (adult body mass [g], basal metabolic rate [mL O₂/hour], and range size [km²]) for 4999 mammal species (with 3340, 661, and 547 available measurements respectively), as well as the estimated correlation over time for residual patterns (bottom panel) showing the correlation (y-axis) over 185 million years (x-axis) of evolutionary history for mammals (shaded interval shows the 95% confidence interval).

models, for example, when estimating non-local and lagged impacts of habitat changes resulting from diffusive and lagged interactions.

Similarly, there is growing interest in using structural causal models to re-interpret a wide range of ecological analyses (Arif & MacNeil, 2022; Byrnes & Dee, 2025; Grace, 2024). Usefully, GGMMs predict the covariance across space, time, and categories by constructing the path matrix as the sum across structural interaction like spatial diffusion, time lags, and interactions among species. Therefore, GGMMs can be used in the causal modelling workflow, that is, developing a graph from scientific knowledge, testing whether it can be rejected as being inconsistent with available data, and subsequently using it to explore the potential impact of policy changes due to both direct and indirect effects (summarizing fig. 2 from Arif & MacNeil, 2023). Usefully, GGMMs allow us to explicitly formulate (and communicate) scientific assumptions about causal relationships (represented by the path matrix), and how these then result in modelled correlations (the inverse of the precision matrix). Ultimately, I agree with Ives (2022) that ecological understanding at large spatial, temporal, and taxonomic scales requires a synthesis of observational and experimental methods, and hope that GGMMs provide a useful path to integrate causal (SEM) and descriptive (GLMM and GAM) statistics.

In summary, I see GGMMs as a useful avenue to integrate ecological theory (i.e. specific ecological interactions across space, time, and eco-evolutionary variables) with statistical estimation (hierarchical modelling tools). The models can be fitted using a range of existing R-packages (e.g. *tinyVAST*), simulated within R, or fitted to new dynamics using bespoke code in RTMB. By providing a unified framework across ecological and evolutionary analyses, I hope that GGMMs will allow researchers to move more easily between predictive, inferential, and causal analyses.

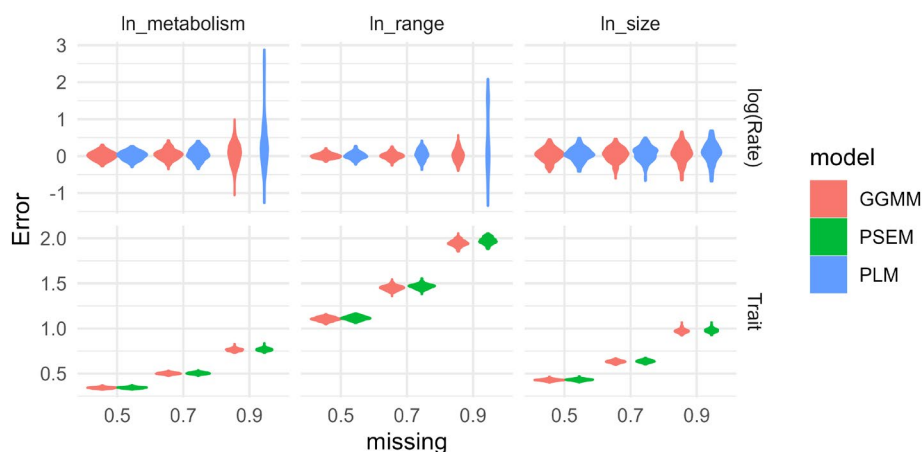


FIGURE 6 Estimation performance in a simulation experiment involving 200 replicates per scenario, where data are simulated based on estimated values from real-world data for the third case study (see Figure 4), comparing performance using the generalized graphical mixed model (GGMM, red), an alternative model that assumes that evolutionary rate is constant among traits (PSEM using package *phylosem*, green), or a model that estimates separate rates but cannot impute missing values during inference (PLM using package *phylolm*, blue). I show the distribution of errors (y-axis) in estimates of the log-Ornstein Uhlenbeck parameter (top row; PSEM is not shown because it incorrectly assumes a constant value across traits) or the average root mean squared error for estimated trait values for each individual taxon (bottom row; PLM is not shown because it does not impute missing values) for three traits (columns) and three simulation scenarios where trait measurements are missing for 50%, 70%, or 90% of species.

AUTHOR CONTRIBUTIONS

James T. Thorson completed all stages of the research and writing.

ACKNOWLEDGEMENTS

I thank Han Wu, who is continuing research related to the diffusion-enhanced model presented here, and Carey McGilliard, who developed the rex sole proportional abundance-at-age explored here. I also thank Chris Wikle and Jonathan Bradley for earlier discussions of multivariate spatio-temporal analysis. The manuscript was improved by comments from Cole Monnahan and Lindsay Campbell, as well as three anonymous reviewers.

FUNDING INFORMATION

The author has no funding information to report.

CONFLICT OF INTEREST STATEMENT

The author has no conflict of interest to report.

DATA AVAILABILITY STATEMENT

All code and data required to replicate analyses and figures are available via GitHub (https://github.com/James-Thorson/graphical_mixed_model/) using release 1.0.0, with DOI provided by Zenodo (<https://zenodo.org/records/18509086>) (Thorson, 2026). The mammal phylogeny was downloaded from VertLife (<https://vertlife.org/phylosubsets/>) and was developed by Upham et al. (2019). The mammal traits were accessed from PanTHERIA (Jones et al., 2009), available online from ESA archives (<https://esapubs.org/archive/ecol/E090/184/metadata.htm>). The proportional abundance-at-age data for rex sole in the Gulf of Alaska is publicly available (https://github.com/noaa-afsc/goa_rex/blob/main/runs/2025_cie_review/2021_accepted_model_inputs/GOA_Rex_8_2021.dat) from the 2024 stock assessment (McGilliard, 2024) and distributed for a Center for Independent Experts 2025 review of the rex sole assessment.

ORCID

James T. Thorson  <https://orcid.org/0000-0001-7415-1010>

REFERENCES

- Arif, S., & MacNeil, M. A. (2022). Predictive models aren't for causal inference. *Ecology Letters*, 25(8), 1741–1745. <https://doi.org/10.1111/ele.14033>
- Arif, S., & MacNeil, M. A. (2023). Applying the structural causal model framework for observational causal inference in ecology. *Ecological Monographs*, 93(1), e1554. <https://doi.org/10.1002/ecm.1554>
- Bates, D., Maechler, M., & Jagan, M. (2023). *Matrix: Sparse and dense matrix classes and methods*. <https://CRAN.R-project.org/package=Matrix>
- Bolker, B. M., Brooks, M. E., Clark, C. J., Geange, S. W., Poulsen, J. R., Stevens, M. H. H., & White, J. S. S. (2009). Generalized linear mixed models: A practical guide for ecology and evolution. *Trends in Ecology & Evolution*, 24(3), 127–135.
- Byrnes, J. E. K., & Dee, L. E. (2025). Causal inference with observational data and unobserved confounding variables. *Ecology Letters*, 28(1), e70023. <https://doi.org/10.1111/ele.70023>
- Foster, S. D., & Bravington, M. V. (2013). A Poisson–Gamma model for analysis of ecological non-negative continuous data. *Environmental and Ecological Statistics*, 20(4), 533–552. <https://doi.org/10.1007/s10651-012-0233-0>
- García-Callejas, D., Lavorel, S., Ovaskainen, O., Peltzer, D. A., & Tylianakis, J. M. (2025). Species traits and landscape structure can drive scale-dependent propagation of effects in ecosystems. *Nature Communications*, 16(1), 7998. <https://doi.org/10.1038/s41467-025-63208-5>
- Grace, J. B. (2006). *Structural equation modeling and natural systems*. Cambridge University Press.
- Grace, J. B. (2024). An integrative paradigm for building causal knowledge. *Ecological Monographs*, 94(4), e1628. <https://doi.org/10.1002/ecm.1628>
- Guennebaud, G., & Jacob, B. (2010). *Eigen3*. <https://libeigen.gitlab.io/>
- Hanski, I., Kuussaari, M., & Nieminen, M. (1994). Metapopulation structure and migration in the butterfly *Melitaea cinxia*. *Ecology*, 75(3), 747–762. <https://doi.org/10.2307/1941732>
- Hautmann, M. (2020). What is macroevolution? *Palaeontology*, 63(1), 1–11. <https://doi.org/10.1111/pala.12465>
- Holt, R. D. (1997). Community modules. *Multitrophic Interactions in Terrestrial Ecosystems, 36th Symposium of the British Ecological Society*, 333–349.
- Ives, A. R. (2022). Random errors are neither: On the interpretation of correlated data. *Methods in Ecology and Evolution*, 13(10), 2092–2105. <https://doi.org/10.1111/2041-210X.13971>
- Ives, A. R., Dennis, B., Cottingham, K. L., & Carpenter, S. R. (2003). Estimating community stability and ecological interactions from time-series data. *Ecological Monographs*, 73(2), 301–330.
- Jetz, W., McGeoch, M. A., Guralnick, R., Ferrier, S., Beck, J., Costello, M. J., Fernandez, M., Geller, G. N., Keil, P., Merow, C., Meyer, C., Muller-Karger, F. E., Pereira, H. M., Regan, E. C., Schmeller, D. S., & Turak, E. (2019). Essential biodiversity variables for mapping and monitoring species populations. *Nature Ecology & Evolution*, 3(4), 539–551. <https://doi.org/10.1038/s41559-019-0826-1>
- Jones, K. E., Bielby, J., Cardillo, M., Fritz, S. A., O'Dell, J., Orme, C. D. L., Safi, K., Sechrest, W., Boakes, E. H., Carbone, C., Connolly, C., Cutts, M. J., Foster, J. K., Grenyer, R., Habib, M., Plaster, C. A., Price, S. A., Rigby, E. A., Rist, J., ... Purvis, A. (2009). PanTHERIA: A species-level database of life history, ecology, and geography of extant and recently extinct mammals. *Ecology*, 90(9), 2648. <https://doi.org/10.1890/08-1494.1>
- Kéry, M., & Kellner, K. F. (2024). *Applied statistical modelling for ecologists: A practical guide to Bayesian and likelihood inference using R, JAGS, NIMBLE, Stan and TMB*. Elsevier.
- Kristensen, K. (2024). RTMB: “R” Bindings for “TMB”. <https://CRAN.R-project.org/package=RTMB>
- Kristensen, K., Nielsen, A., Berg, C. W., Skaug, H., & Bell, B. M. (2016). TMB: Automatic differentiation and Laplace approximation. *Journal of Statistical Software*, 70(5), 1–21. <https://doi.org/10.18637/jss.v070.i05>
- Lande, R. (1976). Natural selection and random genetic drift in phenotypic evolution. *Evolution*, 30, 314–334.
- Leibold, M. A., Holyoak, M., Mouquet, N., Amarasekare, P., Chase, J. M., Hoopes, K. F., Holt, R. D., Shurin, J. B., Law, R., Tilman, D., Loreau, M., & Gonzalez, A. (2004). The metacommunity concept: A framework for multi-scale community ecology. *Ecology Letters*, 7(7), 601–613. <https://doi.org/10.1111/j.1461-0248.2004.00608.x>
- Levin, S. A. (1974). Dispersion and population interactions. *The American Naturalist*, 108(960), 207–228. <https://doi.org/10.1086/282900>
- Levins, R. (1966). The strategy of model building in population biology. *American Scientist*, 54, 421–431.
- Lindgren, F., Bakka, H., Bolin, D., Krainski, E., & Rue, H. (2023). A diffusion-based spatio-temporal extension of Gaussian Matérn fields. (arXiv:2006.04917) arXiv. <https://doi.org/10.48550/arXiv.2006.04917>
- Lindgren, F., Rue, H., & Lindström, J. (2011). An explicit link between Gaussian fields and Gaussian Markov random fields: The stochastic

- partial differential equation approach. *Journal of the Royal Statistical Society. Series B, Statistical Methodology*, 73(4), 423–498. <https://doi.org/10.1111/j.1467-9868.2011.00777.x>
- Lindmark, M., Anderson, S. C., & Thorson, J. T. (2026). Estimating scale-dependent covariate responses using two-dimensional diffusion derived from the stochastic partial differential equation method. *Methods in Ecology and Evolution*, 17(1), 207–218. <https://doi.org/10.1111/2041-210X.70177>
- McGilliard, C. R. (2024). Assessment of the Rex Sole Stock in the Gulf of Alaska. North Pacific Fishery Management Council. <https://www.npfmc.org/wp-content/PDFdocuments/SAFE/2024/GOArex.pdf>
- Miller, D. L. (2025). Bayesian views of generalized additive modelling. *Methods in Ecology and Evolution*, 16(3), 446–455. <https://doi.org/10.1111/2041-210X.14498>
- Rue, H., & Held, L. (2005). *Gaussian Markov random fields: Theory and applications* (1st ed.). CRC Press.
- Schliep, E. M., Lany, N. K., Zarnetske, P. L., Schaeffer, R. N., Orians, C. M., Orwig, D. A., & Preisser, E. L. (2018). Joint species distribution modelling for spatio-temporal occurrence and ordinal abundance data. *Global Ecology and Biogeography*, 27(1), 142–155. <https://doi.org/10.1111/geb.12666>
- Skaug, H., & Fournier, D. (2006). Automatic approximation of the marginal likelihood in non-Gaussian hierarchical models. *Computational Statistics & Data Analysis*, 51(2), 699–709.
- Thorson, J. T. (2026). *James-Thorson-NOAA/graphical_mixed_model: Release for accepted version* [Computer software]. Zenodo. <https://doi.org/10.5281/zenodo.18509086>
- Thorson, J. T., Anderson, S. C., Goddard, P., & Rooper, C. N. (2025). tiny-VAST: R package with an expressive Interface to specify lagged and simultaneous effects in multivariate Spatio-temporal models. *Global Ecology and Biogeography*, 34(4), e70035. <https://doi.org/10.1111/geb.70035>
- Thorson, J. T., Andrews, A. G., III, Essington, T. E., & Large, S. I. (2024). Dynamic structural equation models synthesize ecosystem dynamics constrained by ecological mechanisms. *Methods in Ecology and Evolution*, 15(4), 744–755. <https://doi.org/10.1111/2041-210X.14289>
- Thorson, J. T., Maureaud, A. A., Frelat, R., Mérigot, B., Bigman, J. S., Friedman, S. T., Palomares, M. L. D., Pinsky, M. L., Price, S. A., & Wainwright, P. (2023). Identifying direct and indirect associations among traits by merging phylogenetic comparative methods and structural equation models. *Methods in Ecology and Evolution*, 14(5), 1259–1275. <https://doi.org/10.1111/2041-210X.14076>
- Thorson, J. T., Monnahan, C. C., & Hulson, P.-J. F. (2023). Data weighting: An iterative process linking surveys, data synthesis, and population models to evaluate mis-specification. *Fisheries Research*, 266, 106762. <https://doi.org/10.1016/j.fishres.2023.106762>
- Thorson, J. T., Munch, S. B., & Swain, D. P. (2017). Estimating partial regulation in spatiotemporal models of community dynamics. *Ecology*, 98(5), 1277–1289. <https://doi.org/10.1002/ecy.1760>
- Thorson, J. T., & van der Bijl, W. (2023). phylosem: A fast and simple R package for phylogenetic inference and trait imputation using phylogenetic structural equation models. *Journal of Evolutionary Biology*, 36(10), 1357–1364. <https://doi.org/10.1111/jeb.14234>
- Tobler, W. R. (1970). A computer movie simulating urban growth in the Detroit region. *Economic Geography*, 46, 234–240.
- Tung Ho, L. s., & Ané, C. (2014). A linear-time algorithm for Gaussian and non-Gaussian trait evolution models. *Systematic Biology*, 63(3), 397–408.
- Turchin, P. (2003). *Complex population dynamics: A theoretical/empirical synthesis*. Princeton University Press.
- Turing, A. M. (1952). The chemical basis of morphogenesis. *Philosophical Transactions of the Royal Society of London. Series B, Biological Sciences*, 237(641), 37–72.
- Upham, N. S., Esselstyn, J. A., & Jetz, W. (2019). Inferring the mammal tree: Species-level sets of phylogenies for questions in ecology, evolution, and conservation. *PLoS Biology*, 17(12), e3000494. <https://doi.org/10.1371/journal.pbio.3000494>
- Ver Hoef, J. M., Hanks, E. M., & Hooten, M. B. (2018). On the relationship between conditional (CAR) and simultaneous (SAR) autoregressive models. *Spatial Statistics*, 25, 68–85. <https://doi.org/10.1016/j.spasta.2018.04.006>
- Wood, S. N. (2017). *Generalized additive models: An introduction with R*. CRC Press.
- Wootton, J. T., & Emmerson, M. (2005). Measurement of interaction strength in nature. *Annual Review of Ecology, Evolution, and Systematics*, 36, 419–444.
- Wright, S. (1921). Correlation and causation. *Journal of Agricultural Research*, 20(7), 557–585.

SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

Supporting Information 1. Derivation of a Gaussian Markov random field from a simultaneous equation model.

Supporting Information 2. Model details for case study 1.

Supporting Information 3. Model details for case study 2.

Supporting Information 4. Model details for case study 3.

Supporting Information 5. Additional figures and tables.

Table S1. Count of trait measurements available for each individual trait (along the diagonal) or any pair of traits (off-diagonal), from the 4999 species available in PanTHERIA that can be matched with the Vertlife phylogeny based on scientific binomial.

Table S2. Results from a simple-random 10-fold cross-validation experiment for each of eight models for proportional abundance-at-age for rex sole in the Gulf of Alaska, arising from every combination of estimating three potential interaction parameters or fixing them at zero. For each model, we list the interaction parameters included, the number of fixed effects, the cross-validation root mean squared error, and the proportion of cross-validation mean-squared error relative to the null model. Note that each model includes 22 parameters in addition to the interactions: an intercept for each age $a \in \{2, 3, \dots, 20\}$; the variance for exogenous variation in the graphical model; and the dispersion and power parameters for the Tweedie distribution for residual variation.

Figure S1. Observed (black bullets) and predicted (lines) proportional abundance-at-age (y-axis) for ages 2–20 (x-axis) in each year 1992–2022 (panels) for rex sole in the Gulf of Alaska, showing the prediction using all data (black line) or using a leave-year-out cross-validation design (red line). Note that (1) the age-20 category includes all animals aged 20+, and (2) the red and black lines are identical in years with no data (no black dots, e.g. 1993) because the leave-year-out cross-validation fits to all available data during that model run.

Figure S2. Histograms showing the count of simulation replicates (y-axis) that result in a given z-value ($z = \frac{\hat{\theta} - \theta}{\text{SE}(\hat{\theta})}$, x-axis) for estimates of the natural-logarithm of the Ornstein-Uhlenbeck parameter relative to the known (true) value used to simulate data for each of 200 simulation replicates, when fitting a generalized graphical mixed model. The parameter is estimated separately for each of three

traits (rows), and we explore three scenarios where 50%, 70%, or 90% of taxa are independently missing a measurement for a given trait (columns). A well-performing model will accurately estimate uncertainty, such that the z-value will follow a normal distribution (black line, showing the normal probability density function multiplied by the number of replicates and histogram bin width).

How to cite this article: Thorson, J. T. (2026). Generalized graphical mixed models connect ecological theory with widely used statistical models. *Methods in Ecology and Evolution*, 17, 1290–1302. <https://doi.org/10.1111/2041-210x.70272>